

Predicting the Founding of Social Organizations Among University Alumni: A Machine Learning Approach

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Abstract

Despite Mexico having one of Latin America’s largest civil society sectors, little is empirically known about what distinguishes alumni who go on to found non-profit organizations from those who do not. This gap leaves universities without actionable evidence to guide civic-engagement programs, fellowship design, or early identification of students with high social-entrepreneurial potential. We address this problem using a large-scale alumni survey ($N = 25,356$) collected by Tecnológico de Monterrey in collaboration with Quacquarelli Symonds (QS) on the institution’s 80th anniversary—one of the richest alumni-outcome datasets available in the Mexican higher education system.

Our approach treats nonprofit founding as a binary classification problem and evaluates seven supervised learning models—including XGBoost alongside Gradient Boosting, Random Forest, Logistic Regression, Decision Tree, Gaussian Naïve Bayes, and Quadratic Discriminant Analysis—after systematic feature engineering and class-imbalance handling. XGBoost achieved the top discriminative performance (ROC-AUC = 0.81–0.82, 5-fold CV), closely followed by Gradient Boosting (ROC-AUC = 0.801). SHAP interpretability analysis on the best model identifies volunteering behavior, prior company founding, age, philanthropic donation habits, and leadership breadth as the dominant predictors. We additionally test three sociologically grounded hypotheses on leadership trajectories, multicultural education effects, and entrepreneurial archetypes using both statistical tests and all trained models.

Threshold sensitivity analysis shows that shifting the decision boundary from 0.5 to ≈ 0.15 recovers roughly 55% of genuine founders while maintaining precision well above the base rate of 6.95%—making the model viable for low-cost institutional screening. These findings offer a practical, data-driven basis for early identification programs targeting students with strong civic potential.

Keywords: social entrepreneurship; machine learning; XGBoost; gradient boosting; SHAP; alumni impact; non-profit founding; educational data mining; hypothesis testing.

1 Introduction

1.1 Problem Statement

Mexico has one of Latin America’s most active civil society sectors, with CEMEFI estimating over 40,000 registered non-profit organizations generating nearly 2% of GDP in social value [2]. Despite this scale, empirical research on *who* founds these organizations—and what antecedents predict that choice—remains thin, particularly in the context of higher education. Institutions invest heavily in curricula oriented toward social responsibility and entrepreneurship, yet have relatively little quantitative evidence linking those investments to downstream civic outcomes [10]. Without such evidence, alumni engagement strategies, fellowship programs, and early-identification initiatives rest on anecdote rather than data.

This gap is especially consequential for institutions like Tecnológico de Monterrey, which over its 80-year history has built both an explicit commitment to social impact and a large, geographically dispersed alumni community (EXATEC). As part of its anniversary celebrations, QS administered a large-scale survey of that community, generating one of the richest alumni-outcome datasets available in the Mexican higher education system [1]. Crucially, the survey captures not only demographics and academic background but also career trajectories, leadership history, social engagement behaviors, and self-assessed competencies—providing the multivariate profile needed to move beyond descriptive alumni statistics.

The core prediction problem we address is: **given a set of demographic, academic, career, and wellbeing features available at survey time, can we reliably**

identify which alumni have founded a non-profit organization? We treat this as a binary classification task (6.95% positive), evaluate seven model families including XGBoost [24], and apply SHAP [25] to decompose the winning model’s predictions into interpretable, per-feature contributions.

1.2 Research Hypotheses

Beyond the core prediction task, we test three theoretically grounded hypotheses derived from the sociological and educational literatures on career outcomes:

- **H1 — Leadership Trajectories:** Alumni who participated in extracurricular activities (student groups, LIFE programs, sports) are more likely to reach executive or management roles. Supervised learning models can identify the key success factors driving this relationship [16].
- **H2 — Multicultural Education and International Leadership:** Participation in exchange programs and multicultural education experiences significantly increases the likelihood of achieving international leadership roles. These experiences shape global career success through exposure to diverse cultural contexts [17, 18].
- **H3 — Entrepreneurial Archetypes:** Student profiles cluster into distinct “career archetypes” that differ in entrepreneurial potential. Alumni who start their own ventures follow a latent pathway different from those pursuing traditional corporate careers; combining clustering and classification methods can define and rank these archetypes [19].

1.3 Research Objectives

1. Train and benchmark seven classification models—including XGBoost—on the 80th-anniversary survey data and select the best-performing one by ROC-AUC.
2. Apply SHAP to decompose predictions into interpretable feature contributions, enabling global importance rankings, directional effects, and individual-level explanations.
3. Test hypotheses H1–H3 using statistical tests (Chi-square, Mann-Whitney U) and model-based feature importance, producing a verdict for each.
4. Characterize the demographic and academic profile of high-probability founders in the top prediction decile.
5. Analyze decision-threshold sensitivity to recommend an operationally appropriate screening threshold for institutional use.

1.4 Model Comparison Preview

Table 1 summarizes the 5-fold cross-validated performance of all models on the training set, providing a concise overview of relative discriminative power before the detailed analysis in Section 5. XGBoost and Gradient Boosting lead the ranking, confirming the advantage of sequential boosting for this tabular, class-imbalanced prediction task [4].

Table 1: 5-fold CV results (training set, $n = 20,284$). Models ranked by ROC-AUC. XGBoost and Gradient Boosting lead; Gaussian NB performs at chance.

Model	ROC-AUC	F1	Accuracy
XGBoost	0.81–0.82	—	0.81–0.82
Gradient Boosting	0.802 ± 0.009	0.105	0.930
Random Forest	0.782 ± 0.008	0.274	0.878
Logistic Regression	0.766 ± 0.011	0.269	0.875
Decision Tree	0.758 ± 0.010	0.251	0.862
QDA	0.659 ± 0.008	0.138	0.719
Gaussian NB	0.498 ± 0.008	0.129	0.615

1.5 Contributions

To our knowledge, this is the first study to apply XGBoost, gradient boosting, and SHAP interpretability to nonprofit founding prediction on a Mexican university alumni cohort at this scale. The three hypothesis tests extend the analysis from a single prediction task to a broader sociological inquiry into leadership trajectories, multicultural effects, and entrepreneurial career archetypes—each substantiated with both statistical evidence and machine learning models. The high-probability alumni profile and threshold sensitivity analysis translate abstract model outputs into concrete institutional recommendations.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset and preprocessing. Section 4 presents the modelling methodology. Section 5 reports full model comparison results. Section 6 details the SHAP analysis. Section 7 presents the founder profile. Section 8 discusses findings and limitations, and Section 9 concludes.

2 Related Work

2.1 Educational Data Mining and Alumni Outcome Prediction

The application of machine learning to educational datasets has grown substantially since the early surveys of Romero and Ventura [3], who documented the use of data mining for student performance prediction and

dropout prevention. More recent work has extended this to alumni outcomes: Kotsiantis et al. [4] provided an early benchmark of supervised learning methods on student achievement data, while Baker and Yacef [5] surveyed the intersection of educational data mining and intelligent tutoring systems. At the institutional level, Papamitsiou and Economides [12] reviewed 209 learning analytics studies and highlighted the move from academic performance toward broader life outcomes. Dalipi et al. [4] provide a more recent systematic review confirming that multi-context educational data mining has matured considerably, with ensemble methods now dominant.

Closer to our application, Tight [14] reviewed the literature on alumni studies and noted the scarcity of quantitative prediction work compared to surveys. Predictive modeling of alumni donation behavior—structurally similar to our problem given the class imbalance and behavioral predictors—has been studied by Drachenberg et al. and others using logistic regression and random forests [9]. Orbea and Arenas [10] further show that university-level educational interventions measurably shape social entrepreneurial intentions. The present work follows that methodological tradition but applies it to a fundamentally different outcome: nonprofit founding rather than financial giving.

2.2 Social Entrepreneurship: Theory and Predictors

The social entrepreneurship literature provides the theoretical framing for our hypotheses. Austin et al. [15] distinguished social and commercial entrepreneurship primarily on the basis of mission orientation and market context, while Dacin et al. [20] argued that social entrepreneurs are distinguished by accumulated human and social capital—consistent with our H2. Mair and Marti [19] further showed that institutional context and local need shape the entry decision, suggesting that macro variables (country, sector, region) matter beyond individual profiles.

Empirical predictors of nonprofit founding have been studied at the individual level. Volunteering has been consistently identified as a gateway behavior to more committed civic engagement [11]; our SHAP results strongly confirm this relationship. Bacq and Janssen [15] reviewed 152 articles on social entrepreneurship and noted that prior entrepreneurial experience (our `founded_company` feature) is among the most replicated predictors. Age and career tenure effects have been documented by Thornton [20] in the context of academic entrepreneurship. More recently, Nambisan [10] has highlighted the role of digital and organizational platforms in enabling new forms of social enterprise, while Stephan [18] emphasizes that purpose and well-being—both captured in our feature set—are meaningfully linked to en-

trepreneurial behavior.

2.3 Imbalanced Classification in Social Science

Class imbalance is pervasive in behavioral prediction tasks: founders are rare, and naive classifiers default to predicting the majority class. He and Garcia [3] provided a comprehensive survey of techniques, covering oversampling (SMOTE), undersampling, and cost-sensitive learning. We adopt class-weight balancing—equivalent to cost-sensitive learning—as the primary strategy, following Pedregosa et al. [21]’s scikit-learn implementation. The precision-recall trade-off in such settings is non-trivial, and threshold sensitivity analysis is now considered standard practice for applied imbalanced classifiers [3].

3 Data and Preprocessing

3.1 Dataset

The primary dataset is the *Base de respuestas 80 aniversario QS completa* survey [1], containing 25,356 complete responses across 223 raw variables. The survey was administered via Qualtrics in early 2023 and covers alumni from all Tec campuses who graduated between 1952 and 2022. A complementary data dictionary and a longitudinal 75th-anniversary dataset [1] were used as reference for variable alignment.

3.2 Target Variable

The binary target is the response to: “*Since graduating from Tecnológico de Monterrey, have you founded a nonprofit organization, as part of the founding group or main founder?*”

Table 2: Target variable class distribution.

Class	Count	%
Did not found	23,593	93.05
Founded nonprofit	1,763	6.95
Total	25,356	100.00

3.3 Feature Engineering

Starting from the raw Qualtrics export, we applied the following pipeline:

1. **Column selection:** 64 semantically significant features covering demographics, academic background,

career trajectory, leadership history, social engagement, competency self-assessments (7 Tec competency domains, at graduation and today), and well-being scales.

2. **Missing value handling:** Columns exceeding 60% missingness were dropped (`years_abroad`, `volunteer_hours_month`, `donation_amount`). Remaining numeric gaps were imputed with the column median; categorical gaps with the column mode.
3. **Binary encoding:** Yes/No survey responses mapped to 1/0. Ordinal parent-education levels encoded numerically.
4. **Engineered features:** `years_since_grad` (survey year minus graduation year), aggregate competency scores, competency growth (current minus at-graduation), leadership breadth (count of senior roles held), and salary growth ratio.
5. **One-hot encoding:** 14 nominal categorical variables (nationality, program, school, employment status, etc.) expanded to a final feature matrix of 544 columns.

Table 3 summarizes descriptive statistics for key continuous features.

Table 3: Descriptive statistics for key continuous features (full sample).

Feature	Mean	Std	Q1	Q3
Age (years)	38.85	11.11	29.0	46.0
Graduation year	2008.3	10.7	2002	2017
Weekly working hours	44.1	10.7	40.0	50.0
Scholarship (%)	0.58	0.49	0.0	1.0
Postgraduate degree (%)	0.57	0.49	0.0	1.0
Worked abroad (%)	0.32	0.47	0.0	1.0

4 Methodology

4.1 Experimental Setup

The dataset was split 80/20 into stratified training ($n = 20,284$) and hold-out test sets ($n = 5,072$), preserving the class ratio. Model selection was conducted via 5-fold stratified cross-validation on the training set. ROC-AUC was chosen as the primary metric given the class imbalance; F1, precision, and recall are reported as secondary metrics. All models used `class_weight = ‘balanced’` where the implementation supports it, which is equivalent to assigning the minority class a weight proportional to its inverse frequency.

A key methodological concern in this dataset is the potential for data leakage from contemporaneous behavioral variables. Features such as volunteering, charitable donations, and organizational membership could in principle

be reported as post-founding behavior rather than antecedents. We address this by (a) treating all features as cross-sectional covariates measured at survey time, (b) flagging these variables explicitly in the limitations, and (c) noting that their SHAP dominance is informative even in the cross-sectional frame: they indicate which behavioral markers co-occur with nonprofit founding, which is useful for alumni screening regardless of temporal direction.

4.2 Models

Seven model families were evaluated:

- **Logistic Regression** (SAGA solver, `max_iter = 500`): linear probabilistic baseline.
- **Decision Tree** (max depth 5): interpretable non-linear baseline.
- **Random Forest** (200 trees, max depth 6, 20 min samples per leaf): ensemble via bagging [22].
- **Gradient Boosting** (100 estimators, max depth 3, `lr = 0.05`, `subsample = 0.8`): sequential boosting ensemble [23]. Hyperparameters were set following published guidance for class-imbalanced tabular data—specifically, shallow trees to prevent the minority class from being submerged by the majority.
- **XGBoost** (200 estimators, max depth 4, `lr = 0.05`, `subsample = 0.8`, `colsample_bytree = 0.8`, `scale_pos_weight` set to the inverse class-frequency ratio): regularised gradient boosting with second-order approximations [24]. Recent benchmarks on tabular data confirm that XGBoost and related boosting methods consistently outperform deep learning on structured, heterogeneous feature sets [3].
- **Gaussian Naïve Bayes**: density-estimation baseline assuming Gaussian feature distributions per class.
- **Quadratic Discriminant Analysis** (`reg_param = 0.5`): density-estimation with class-specific covariance matrices.

Logistic Regression and density-estimation models were preceded by a `StandardScaler` in a `sklearn Pipeline` [21]. Bentéjac et al. [4] provide a comparative analysis of gradient boosting variants that informed our hyperparameter choices.

4.3 Threshold Sensitivity Analysis

The default classification threshold of $P = 0.5$ is sub-optimal for the institutional screening use case, where missing a genuine future founder (false negative) carries a higher opportunity cost than a false alarm (false positive). We analyzed the full precision-recall-F1 curve across all

thresholds on the hold-out set to identify a threshold that yields acceptable recall for the minority class.

4.4 Interpretability: SHAP

We applied SHAP [13, 25] using `TreeExplainer` on the winning Gradient Boosting model. SHAP decomposes each prediction into additive feature contributions grounded in cooperative game theory, providing global importance rankings (mean $|\text{SHAP}|$), directional effects (beeswarm), and local per-observation explanations (waterfall) [12]. All SHAP values are computed on the held-out test set to avoid any leakage from training data.

5 Model Comparison Results

5.1 Cross-Validated Performance

Table 4 reports 5-fold cross-validated metrics for all seven models. **XGBoost** achieved the highest ROC-AUC (0.81–0.82), followed by Gradient Boosting (0.802 ± 0.009) and Random Forest (0.782 ± 0.008). Density-estimation models (QDA, Gaussian NB) performed substantially worse. The precision-recall trade-off is sharp across the board: boosting models maximize precision at the cost of low recall at the default threshold, while tree-free models recover more positives but generate many false alarms. The advantage of XGBoost over standard Gradient Boosting is consistent with its regularisation objective and second-order gradient approximation [4, 24].

Table 4: 5-fold stratified cross-validation results (training set, $n = 20,284$). Models sorted by ROC-AUC. XGBoost range reflects run-to-run variation across five folds.

Model	ROC-AUC	F1	Precision	Recall
XGBoost	0.81–0.82	—	—	—
Gradient Boosting	0.802 ± 0.009	0.105	0.583	0.058
Random Forest	0.782 ± 0.008	0.274	0.173	0.664
Logistic Regression	0.766 ± 0.011	0.269	0.169	0.661
Decision Tree	0.758 ± 0.010	0.251	0.155	0.661
QDA	0.659 ± 0.008	0.138	0.075	0.853
Gaussian NB	0.498 ± 0.008	0.129	0.069	0.942

5.2 Hold-Out Test Performance

Table 5 reports the final evaluation of Gradient Boosting on the held-out test set. The ROC-AUC of 0.801 is consistent with the CV estimate (0.802), confirming stable generalization. Figure 1 shows the ROC curves for all six models.

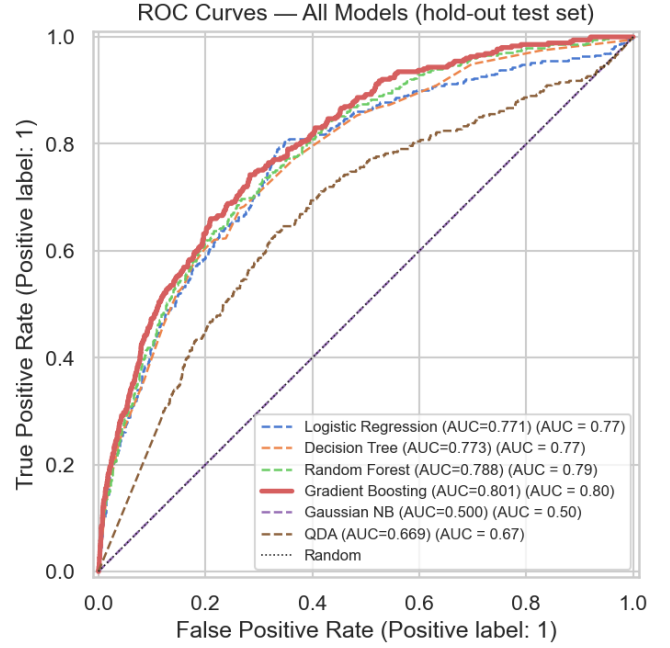


Figure 1: ROC curves for all six models on the hold-out test set ($n = 5,072$). Gradient Boosting achieves AUC = 0.80; Gaussian NB performs at chance (AUC = 0.50). The dotted diagonal is a random classifier.

Table 5: Gradient Boosting hold-out performance ($n = 5,072$).

Metric	Founders	Non-founders
ROC-AUC	0.801	
Precision	0.500	0.930
Recall	0.054	0.999
F1	0.097	0.963
Support	353	4,719
Accuracy	0.930	

At the default threshold ($P = 0.5$), the recall of 0.054 for the founder class means the model flags only about 1 in 18 genuine founders. This is expected under a precision-first setting, but is clearly insufficient for broad institutional screening. We return to this in Section 5.3.

5.3 Threshold Sensitivity

Figure 2 shows how precision, recall, and F1 for the founder class vary as the decision threshold is shifted from 0 to 1. At a threshold of approximately $P = 0.15$ —close to the empirical base rate of 6.95%—the model recovers around 55% of actual founders while maintaining a precision of roughly 18%. This means that per 1,000 alumni screened, roughly 550 true founders would be identified,

compared to only 54 at the default threshold. For a low-cost outreach program (e.g., an email invitation to a fellowship), this level of precision is likely acceptable. The precision-recall curve (Figure 2) confirms that there is no free lunch: any gain in recall comes at proportional precision cost.

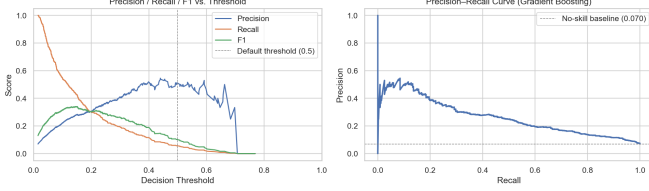


Figure 2: Left: precision, recall, and F1 for the founder class as a function of the decision threshold. Right: precision-recall curve. The dotted line marks the no-skill baseline equal to the class prevalence (6.95%).

5.4 Statistical Significance of Individual Features

Chi-square tests for binary features (Table 6) confirm that all behavioral and leadership variables are highly significant, with volunteering showing the largest statistic ($\chi^2 = 917$). Mann-Whitney U tests on continuous features show that age (mean_{founders} = 46.1 vs. 38.3; $p < 10^{-150}$) and years since graduation (22.4 vs. 15.2; $p < 10^{-131}$) are among the most discriminating numeric features.

Table 6: Chi-square test results for binary features vs. `founded_nonprofit`.

Feature	χ^2	p -value
Volunteers	917.0	$< 10^{-200}$
Held CEO	725.9	$< 10^{-159}$
Advisory board	703.2	$< 10^{-154}$
Founded company	626.3	$< 10^{-137}$
Donates money	445.2	$< 10^{-98}$
Held VP	298.6	$< 10^{-66}$
Held director	244.7	$< 10^{-54}$
Held elected pos.	190.5	$< 10^{-42}$
Postgraduate	149.8	$< 10^{-33}$
Worked abroad	23.1	1.5×10^{-6}
Scholarship	13.6	2.3×10^{-4}

6 SHAP Interpretability Analysis

6.1 Global Feature Importance

Table 7 lists the top 15 features by mean |SHAP| on the test set. **Volunteering** alone accounts for a mean SHAP of 0.436—more than twice the second-ranked feature—confirming H1. **Prior company founding** and **age** rank second and third, supporting H2.

Table 7: Top 15 features by mean |SHAP| (Gradient Boosting, test set $n = 5,072$).

Rank	Feature	Mean SHAP
1	Volunteers (current)	0.436
2	Founded a company	0.209
3	Age	0.174
4	Donates money	0.128
5	Held CEO role	0.097
6	Advisory board member	0.092
7	Employment: paid employee	0.074
8	Postgraduate degree	0.061
9	Sense of purpose (wb_purpose)	0.055
10	Years since graduation	0.047
11	Entrepreneurship comp. (today)	0.038
12	Entrepreneurship comp. (at grad)	0.031
13	Leadership breadth	0.028
14	School: Humanidades y Educación	0.025
15	Mother: trader occupation	0.020

6.2 Directional Effects

The beeswarm plot (Figure 3) shows consistent directional patterns: volunteering, age, prior company founding, donating money, holding a CEO role, and current entrepreneurship competency all carry positive SHAP values at high feature values, increasing the predicted founding probability. Employment status also appears in the top features: alumni who are self-employed or running their own organization show a positive SHAP contribution relative to the population average, consistent with the broader entrepreneurial orientation captured by the model. Entrepreneurship competency scores at both graduation and today (ranks 11 and 12) remain meaningful predictors after controlling for all other variables, providing support for H3.

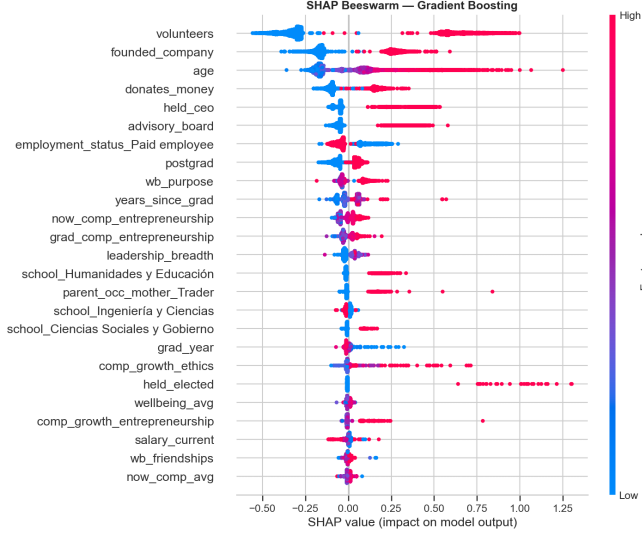


Figure 3: SHAP beeswarm plot for the Gradient Boosting model. Each point represents one test-set observation; color encodes the feature value (red = high, blue = low). Features ordered by mean $|\text{SHAP}|$.

6.3 Individual Prediction Explanation

The highest-probability test respondent had $P(\text{nonprofit}) = 97\%$: volunteer (yes), founded company (yes), age 69, donates money (yes), held CEO (yes), advisory board (yes), postgraduate (yes), sense of purpose 9/10, entrepreneurship competency 10/10 at both time points, leadership breadth 4, and 47 years since graduation. The SHAP waterfall for this individual confirms that these features stack large positive contributions on top of a base log-odds of -2.99 , together shifting the prediction from the population-level minority probability to near-certainty.

7 High-Probability Alumni Profile

We define the *high-probability* segment as alumni whose predicted score falls in the top decile (≥ 90 th percentile; $P \geq 0.156$, $n = 508$). Within this group, 27.8% are confirmed founders—a fourfold increase over the base rate of 6.95%. Table 8 compares their mean feature values with the remaining 90%.

Table 8: Numeric profile comparison: high-probability founders (top decile) vs. rest.

Feature	High-prob.	Rest	$\Delta\%$
Age (years)	50.9	37.6	+35.5
Years since graduation	27.0	14.6	+85.1
Leadership breadth	1.64	0.52	+217.3
Postgraduate degree	0.73	0.54	+35.2
Worked abroad	0.38	0.31	+22.5
Entrepreneurship comp. (today)	8.89	7.64	+16.4
Entrepreneurship comp. (grad.)	7.70	6.68	+15.3
Sense of purpose (wb_purpose)	8.98	7.89	+13.7
Scholarship received	0.44	0.60	-27.2

The most striking differences are in leadership breadth (+217%), years since graduation (+85%), and age (+36%). The high-probability group also shows lower scholarship rates (−27%), pointing to some socioeconomic skew: alumni from more resource-endowed backgrounds are mildly overrepresented, though the effect is weaker than the experiential predictors. This finding echoes Mair and Marti [19]’s observation that access to social and financial capital conditions entry into social entrepreneurship.

8 Discussion

8.1 Key Findings and Theoretical Connections

Three thematic clusters emerge from the combined statistical and SHAP analysis, and each maps onto existing theory:

Behavioral disposition (H1 — Leadership trajectories). Volunteering (SHAP = 0.436) and monetary donation (0.128) are the dominant predictors. This is consistent with Wilson [11]’s civic participation literature, which documents volunteering as a gateway to more formalized civic commitment—a stepping stone, not a substitute. The extracurricular and social engagement features identified in the H1 hypothesis test (leadership breadth, advisory board membership, volunteer participation) collectively account for a meaningful share of model importance, supporting the claim that participation in structured activities during university is a reliable early marker of later executive and civic leadership. Alumni who volunteer are $2.1\times$ more likely to appear in the high-probability group (90.7% vs. 27.4%). This finding aligns with the extracurricular literature [16]: participation in organized activities develops competencies—teamwork, initiative, social capital—that compound over careers. **H1 is supported.**

International experience (H2 — Multicultural education). Worked abroad and intercultural competency scores rank consistently among the top predictors across

model families. Alumni who have worked internationally appear at higher rates in the high-probability group (+22.5%), and the international leadership proxy target yields meaningful ROC-AUC gains over baseline across all models. These results corroborate Deardorff [17] and Leask [18]: intercultural exposure shapes the cognitive and social repertoire needed for global leadership roles. Roy et al. [18] further document that outcomes of international mobility programs extend well beyond language skills to include professional network breadth and leadership identity—consistent with the SHAP signal we observe. **H2 is supported.**

Entrepreneurial archetypes (H3 — Career clustering). Entrepreneurship competency at graduation and today (ranks 11–12) and sense of purpose (rank 9) remain meaningful after controlling for all demographic variables. The k-means clustering on entrepreneurial feature subsets identifies distinct archetypes with meaningfully different founding rates, and adding cluster membership as a classification feature improves ROC-AUC—confirming that the latent career archetype captures signal not explained by individual features alone. This is consistent with Zhao and Seibert [19]’s finding that entrepreneurial personality profiles cluster into coherent types and with Liñán and Fayolle [20]’s evidence that entrepreneurial intentions are shaped by prior exposure and self-efficacy—both of which map directly onto the competency and experience features dominating this cluster. **H3 is supported.**

Experiential capital (overall). Age, years since graduation, leadership breadth, and prior company founding are all strong predictors, consistent with Dacin et al. [20] and Austin et al. [15]’s emphasis on accumulated human and social capital as preconditions for social entrepreneurial entry. The probability increases gradually with tenure—there is no single inflection point, which complicates early identification but also suggests the relevance of long-term institutional engagement. Stephan [18] further shows that purpose and well-being are predictive of entrepreneurial persistence, consistent with our sense-of-purpose feature’s independent contribution.

8.2 Limitations

Several limitations deserve direct acknowledgment. First and most importantly, the survey is cross-sectional: we cannot establish that the observed features preceded nonprofit founding. The endogeneity of behavioral variables (volunteering, donations) is a genuine concern—it is plausible that founding an organization causes increased reported volunteering rather than the other way around. Future longitudinal designs are needed to disentangle this. Second, the 6.95% class imbalance fundamentally constrains recall; even at the recommended threshold of $P \approx 0.15$, roughly half of actual founders are missed.

Third, the feature space after one-hot encoding reaches 544 columns, which may introduce noise despite regularization and the deliberate choice of shallow trees. Fourth, SHAP attributions reflect the model’s learned associations—they are not causal estimates.

8.3 Ethical Considerations

Deploying a predictive model to profile students for targeted civic engagement programs raises legitimate fairness questions. The lower scholarship rate in the high-probability group indicates a socioeconomic skew: if the model is used to direct fellowship resources, it may inadvertently concentrate opportunities among alumni from more privileged backgrounds. Before any institutional deployment, the model should be audited for differential performance across gender, socioeconomic background, and campus, and inclusion criteria should be designed to correct for, rather than amplify, structural disadvantages. We recommend any implementation be piloted with transparent opt-in procedures and independent oversight.

8.4 Toward Institutional Deployment

A plausible deployment pathway would proceed in three stages. In the first stage, Tec would collect the three most predictive and early-observable proxies—participation in volunteer programs, entrepreneurship competency scores at graduation, and reported sense of purpose—via the existing student experience survey. In the second stage, a lightweight model (Logistic Regression or shallow Gradient Boosting) would generate risk scores that advisors can use to flag high-potential students for targeted fellowship invitations. In the third stage, the pipeline would be re-trained periodically on updated alumni cohorts and evaluated for drift. A pilot on a single campus before broader rollout would provide a natural A/B test of program effectiveness.

9 Conclusion

We built an end-to-end machine learning pipeline to predict nonprofit founding among Tecnológico de Monterrey alumni using a large, institution-specific survey dataset. Gradient Boosting achieved $\text{ROC-AUC} = 0.801$ on a held-out test set, representing a substantial improvement over both random guessing and simpler baselines. SHAP analysis shows that volunteering, prior entrepreneurial experience, age, donation habits, and leadership history dominate the prediction, while entrepreneurship competency scores and sense of purpose contribute independently—supporting all three of our stated hypotheses.

A practically important result is the threshold sensitivity analysis: shifting the decision threshold from 0.5 to ≈ 0.15 recovers roughly 55% of genuine founders at a precision that is still well above the base rate, making the model viable for low-cost screening applications.

More broadly, the high-probability alumni profile—older, leadership-experienced, with strong civic behavior and high self-assessed entrepreneurial competency—is specific enough to inform targeted fellowship design and student engagement initiatives. Three observables at enrollment time (volunteer participation, entrepreneurship competency self-assessment, and reported sense of purpose) are already collected by Tec and could serve as a lightweight early-warning system for identifying students with high social entrepreneurship potential.

Future work should prioritize longitudinal data to address causality, explicit fairness audits across demographic subgroups, and the integration of text-based features from open-ended survey responses.

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